

Final Traffic Intelligence Portfolio Capstone Report

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Course: NUS Graduate Capstone Project
Dataset: Metro Interstate Traffic Volume Corridor (48,187 Cleaned Records)

1. Project Executive Summary

This comprehensive portfolio report synthesizes the methodology, findings, and technical components developed to build an end-to-end traffic intelligence solution for the I-94 westbound corridor. By transitioning from initial descriptive data analytics to robust pipeline construction, supervised learning, time-series forecasting, and model quantization, this project provides a scalable framework to optimize regional mobility and manage traffic congestion.

2. Pipeline Methodology and Feature Engineering

The data pipeline constructed handles raw data ingestion via strict validation rules. Sensor anomalies—such as physically impossible 0 Kelvin temperature entries and extreme rainfall outliers—are programmatically caught and imputed utilizing dynamic monthly medians rather than global averages, eliminating data bias. Duplicate records were identified and dropped, resulting in a finalized baseline dataset of 48,187 records. Feature engineering generated downstream model inputs, including Min-Max scaled numerical representations of continuous variables, one-hot encoded weather categories, and cyclical mathematical transformations (sine and cosine components) of temporal dimensions to capture daily and weekly variations smoothly.

3. Model Benchmarking and Performance Metrics

Predictive performance was rigorously audited across multiple machine learning tasks utilizing separate validation partitions:

Modeling Task / Algorithm	Evaluation Metric	Performance Score	Operational Assessment
Classification: Logistic Regression	ROC AUC	0.9910	Solid baseline model performance.
Classification: Random Forest Ensemble	F1-Score	0.9548	Highly precise asset for risk mitigation.
Regression: Linear Model Baseline	Mean Absolute Error	834.28 vehicles	Struggles with non-linear peaks.
Regression: Random Forest Regressor	R-squared (R ²)	0.9433	Explains 94.33% of corridor variance.

Time-Series: Deep Learning LSTM	R-squared (R ²)	0.9484	Captures long-term sequential patterns.
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4. Unsupervised Mining Insights

K-Means clustering separated the traffic records into three distinct operating groups: a Late-Night Low Volume profile (mean volume of 828 vehicles) and two separate Mid-Day Moderate traffic profiles distinguished by cloud cover metrics. Conditional probability and native association rule mining established critical traffic triggers, demonstrating that a weekday journey during morning rush hour windows coinciding with an active thunderstorm or severe drizzle generates a traffic congestion lift multiplier of 3.59x with an empirical confidence level of 89.74%.

5. Route Timing Recommendations and Deployment Simulation

The recommendation engine translates historical averages into clear travel windows. For instance, during weekday rain conditions, the system advises scheduling travel between 02:00 and 03:00 to leverage lower averages (299 vehicles). To support enterprise deployment, a mock API server was constructed to handle JSON requests, returning real-time volume predictions. MLOps safeguards include an automated drift monitoring system that calculates streaming feature distribution shifts, showing a 2.14% variance under normal parameters (PASS) and triggering a high-priority system alert if signal drift breaches the 15% operational tolerance threshold.

6. Actionable Mobility Recommendations

- **Variable Signage Integration:** Link the Random Forest risk classifier directly to digital roadside displays. When high-risk conditions are triggered during wet weather windows, preemptively lower speed limits to smooth traffic flows.
- **Transit Schedule Optimization:** Utilize the deep learning LSTM model's low error bounds to optimize municipal maintenance scheduling, keeping heavy service vehicles off the corridor during projected peak windows.
- **Emergency Asset Allocation:** Pre-position emergency service vehicles near historically verified bottleneck sectors when morning rush hours intersect with adverse weather forecasts, shortening incident response times.